

Venkata Ramana

Irving, TX

+1-214-225-5398 bvenkata188@gmail.com [in](#) [LinkedIn](#)

Summary

Senior Data Engineer with extensive experience designing and implementing scalable ETL pipelines and data solutions across multi-cloud environments. Demonstrated success in reducing processing times and enhancing data governance for high-volume financial and healthcare systems. Proven expertise in Python, Scala, SQL, Spark, and modern data frameworks, collaborating effectively with cross-functional teams to deliver impactful analytics and real-time streaming solutions.

Skills

- **Programming Languages:** Java, Python, Scala, R, SAS, SQL, T-SQL, PL/SQL, JavaScript, TypeScript
- **Frontend Technologies:** HTML, CSS, jQuery
- **Framework & Libraries:** Spring Boot, AngularJS, ReactJS, Node.js
- **Big Data Ecosystem:** Talend, Hadoop, MapReduce, Hive, Apache Spark, Pig, Sqoop, Pyspark, Databricks, Snowflake, Columnar Storage Formats
- **ETL Tools:** Apache NiFi, Apache Kafka, Talend, Apache Airflow, Informatica, DBT, SSRS, SSIS, QlikView, Crystal Reports
- **Cloud Platforms:** AWS (S3, EC2, RDS, Lambda, Redshift), Azure (Blob Storage, Virtual Machines, AI), Google Cloud Platform (GCP)
- **Databases:** MySQL, SQL Server, HBase, Cassandra, DynamoDB, MongoDB, PostgreSQL
- **Visualization Tools:** Tableau, Power BI, Microsoft Excel
- **Development Tools/ DevOps Tools/ Version Control:** JIRA, CI/CD (Jenkins, Azure DevOps), Docker, Kubernetes, Visual Studio, PyCharm, Sublime Text, Eclipse, IntelliJ IDEA, Docker, Git (GitHub, GitLab), DevOps Practices
- **Testing & Performance:** JUnit, Mockito, JMeter, New Relic
- **Packages:** NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn
- **Operating Systems:** Windows, macOS, Linux
- **Functional Skills:** Data Analysis, Data Pipelining, Data Visualization, Data Streaming, Data Integration, Data Transformation, ETL/ELT Processes, Data Modeling, Data Architecture, Data Governance, Compliance, Data Warehouses

Experience

Wells Fargo

Feb 2024 - Present

Data Engineer

USA

- Designed and implemented scalable, end-to-end data pipelines to process 50T+ of financial transactions, market feeds, and risk analytics data, ensuring real-time fraud detection and robust reporting.
- Developed ETL processes ingesting 5M+ daily transactions from diverse financial institutions, reducing data latency and boosting operational efficiency.
- Engineered high-performance data processing workflows using Apache Spark and Hadoop, and deployed data solutions across AWS Redshift, Azure Data Lake, and GCP BigQuery to support scalable financial analytics.
- Constructed real-time and batch ETL pipelines leveraging Apache NiFi, Kafka, Airflow, and DBT, facilitating seamless integration of high-velocity financial transactions across trading platforms.
- Utilized Python for financial modeling, data wrangling, and machine learning workflows, enhancing credit risk prediction accuracy by 30% for advanced fraud detection and algorithmic trading.
- Optimized SQL Server, PostgreSQL, MongoDB, HBase, and Cassandra to enhance query performance by 35%, ensuring low-latency access to critical financial records while upholding transactional integrity.
- Implemented enterprise-grade data governance and security policies using role-based access control, encryption protocols, and secure API integrations, ensuring adherence to PCI-DSS, GDPR, and SOX compliance.
- Collaborated closely with data scientists, risk analysts, and financial stakeholders to deploy AI-powered financial applications, reducing fraudulent transactions by 20%.
- Developed and maintained 45+ Power BI and Tableau dashboards providing real-time visibility into portfolio performance, credit risk exposure, and trading anomalies.
- Streamlined CI/CD pipelines with Docker, Kubernetes, Jenkins, and Azure DevOps, facilitating seamless deployment of cloud-native financial applications.
- Monitored and tuned real-time data performance using JMeter and New Relic, reducing query execution times by 40% while ensuring fault tolerance for mission-critical systems.

Centene Corporation

Nov 2022 - Jan 2024

Data Engineer

USA

- Designed and implemented robust data pipelines for large-scale healthcare datasets, integrating EHRs, insurance claims, and IoT-generated vitals to optimize clinical decision-making.
- Developed both real-time and batch ETL pipelines with Talend, Apache Spark, Hadoop, and SQL, ensuring efficient ingestion and transformation of structured and unstructured data across AWS RDS, Snowflake, Oracle, and Cassandra.

- Engineered scalable data warehousing solutions on AWS platforms to support secure, high-availability storage of FHIR and HL7-compliant patient records while enhancing query performance.
- Leveraged predictive modeling and AI to build ML-driven applications for patient risk scoring and fraud detection, significantly reducing fraudulent claims.
- Implemented full-stack healthcare analytics solutions using ReactJS, AngularJS, and Node.js to create interactive dashboards for real-time monitoring of patient admissions, readmissions, and resource utilization.
- Optimized data processing frameworks with JMeter and New Relic, reducing query execution time by 35% and ensuring high availability and fault tolerance.
- Collaborated with clinicians, business stakeholders, and data scientists in agile environments (Scrum, Jira) to implement AI-powered healthcare solutions, bolstering patient care and operational efficiency.

Larsen & Toubro Infotech

Jan 2020 - Dec 2021

Junior Data Engineer

India

- Engineered and managed petabyte-scale data lakes along with real-time streaming pipelines; ensured seamless integration of 50+ structured and unstructured enterprise data sources, enhancing analytics capabilities for critical business insights.
- Established serverless data pipelines across AWS, Azure, and GCP, achieving a significant reduction in manual data handling tasks, saving the data engineering team over 15 hours per week on routine data operations.
- Automated data workflows with Apache NiFi, Kafka, and Airflow, reducing ETL latency by 40% and ensuring efficient data ingestion from diverse sources.
- Analyzed interactive dashboards and models using Tableau, Power BI, and Snowflake, leading to improved data-driven decision-making and enhanced operational efficiency.
- Integrated predictive analytics pipelines using Spark ML and TensorFlow, optimizing forecasting models and automating data enrichment.
- Implemented advanced encryption techniques within existing data frameworks, reducing unauthorized access incidents to fewer than five per quarter.

Education & Certificates

Trine University

Masters, Information Science and Studies

Angola, IN, USA

KL University

Bachelors, Computer Science and Engineering

Andhra Pradesh, India